RECAP

- If you are a start-up company, which model is more favorable and why?
  - CAPEX or OPEX?

- In virtualization, which characteristic ensures that one virtual machine crashing does not impact others?
  - Partitioning
  - Isolation ✅
  - Encapsulation
  - Hardware Independence

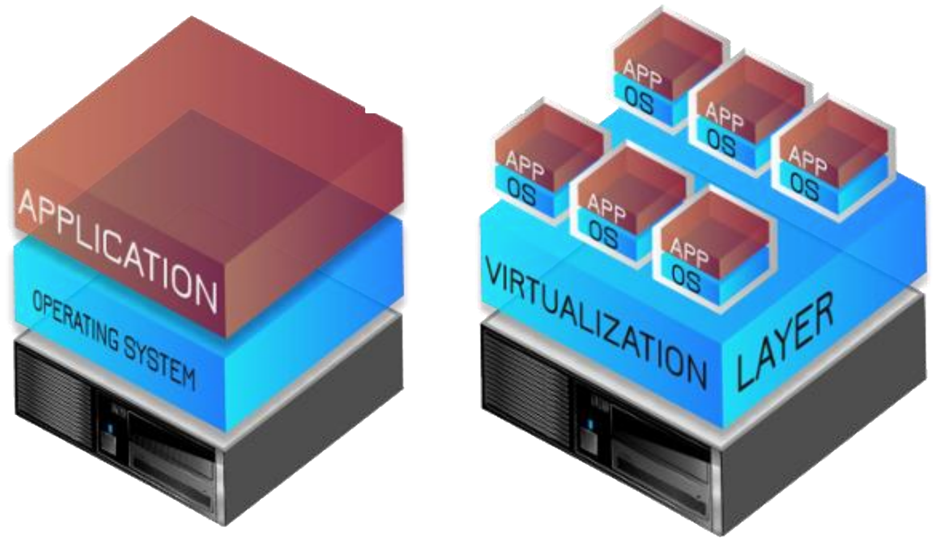
- Overprovisioned workloads moved from on-premise data centers to the cloud will always reduce costs
  - True
  - False ✅

RECAP

- Cloud computing has several economic motivations:
  1. Utility Pricing
     - Cloud service don't need to be cheaper to be economic
     - Whether a solution is economic depends on how to service the demand
  2. Reducing capital expenditures, increasing operational expenses
     - CAPEX requires a large up-front investment to be amortized (written down) as its value decreases over time
     - OPEX is a day-to-day expense which can much more easily be increased or decreased as business needs dictate
  3. Efficiently allocating resources
     - Cloud computing provides the ability to more efficiently allocate compute and storage resources vs dedicated server approaches
     - This translates into more direct control over how business use cloud services that can result in more efficient business operations and ultimately more costs savings

RECAP

- Virtualization
  - Refers to the act of creating a virtual (rather than actual) version of something, including virtual computer hardware platforms, storage devices, and computer network resources
  - The virtualization layer is referred to as a hypervisor, which is a program that allows multiple operating systems to share a single hardware host
  - Two major types: Hosted and Native
  - Advantages:
    - More efficient use of hardware, provisioning fast and simpler, easier to maintain
  - Disadvantages:
    - Virtualization software, licensing, security must protect both physical and virtual machines



Traditional Server Architecture

Virtualized Server Architecture

AWS Overview

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

AWS Console

The ***AWS Management Console*** *is a web application that comprises and refers to a broad collection of service consoles for managing AWS resources. When you first sign in, you see the console home page. The home page provides access to each service console and offers a single place to access the information you need to perform your AWS related tasks*

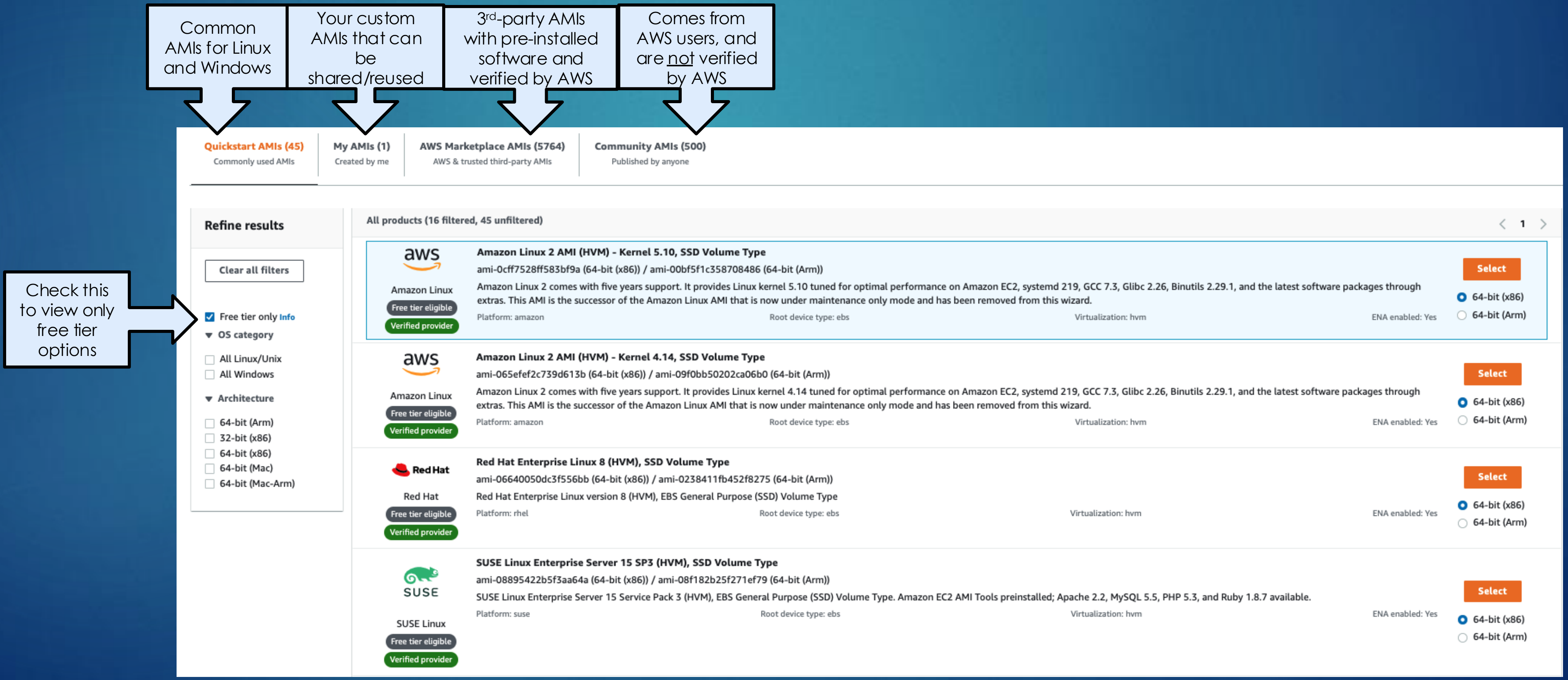
- Over 200 services available
- Customizable interface (e.g. shortcuts)
- Build and monitor applications
- Monitor costs and set alerts
- Aside from a web browser, you can also access it through the mobile app version
- Let's take a tour…

Last class you created a virtual server…

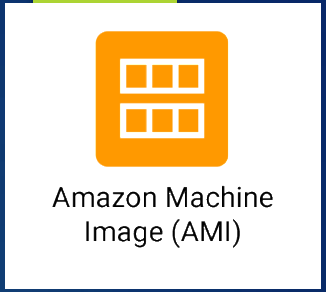
- EC2 (Amazon's Elastic Cloud Computing)
  - Provides the ability for users to create virtual machines as needed (elastic)
  - Users are billed calculated by either the hour or the second based on the size of the instance, operating system, and the Region
  - If you STOP an instance, you will not be charged while it is in a STOPPED state BUT you will for any attached storage since it is persisted
  - The larger the server, the more cost per/hour
- There are several options you can choose when you create an EC2 instance
  - Let's walkthrough them…

Amazon Machine Image (AMI)

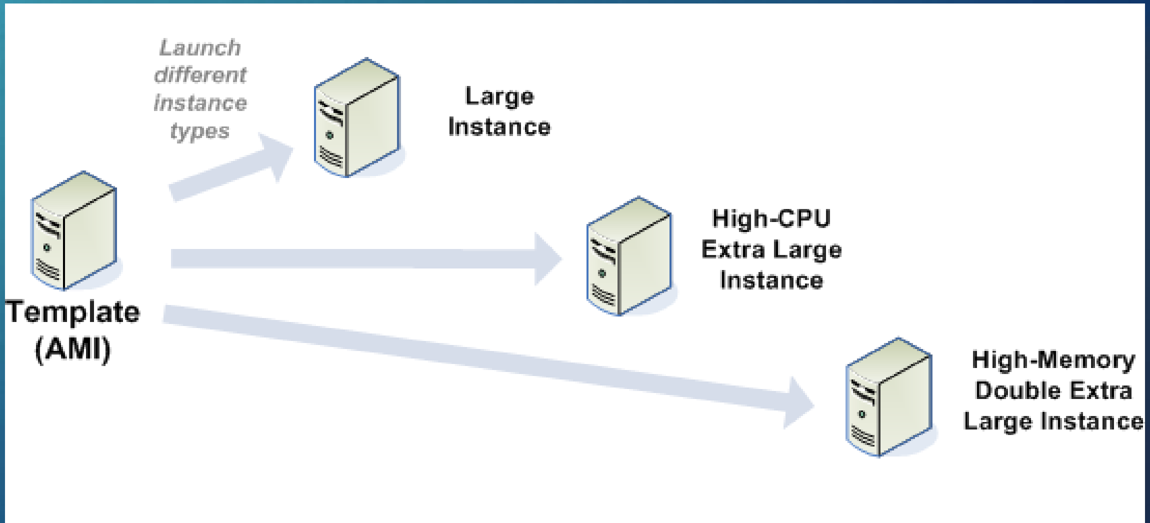
- Virtual machines are created from templates known as an Amazon Machine Instances (AMI)
- It provides the information required to launch an instance



Amazon Machine Image - Details



- An AMI includes the following:
  - A <u>template</u> for the root volume for the instance (for example, an operating system, an application server, and applications)
  - <u>Launch permissions</u> that control which AWS accounts can use the AMI to launch instances
  - A <u>block device mapping</u> that specifies the volumes to attach to the instance when it's launched
- 3 types:
  1. Public – Available to anyone
  2. Paid – Purchase from a developer on AWS Marketplace
  3. Shared – Private, restricted to specific users



Amazon Machine Image – Many Instance configurations to choose from

- From small virtual servers…

	Instance type	vCPUs	Architecture	Memory (GiB)	Storage (GB)	Storage type	Network performance	On-Demand Linux pricing	On-Demand Windows pricing
	t1.micro	1	i386, x86_64	0.612	-	-	Very Low	0.02 USD per Hour	0.02 USD per Hour
	t2.nano	1	i386, x86_64	0.5	-	-	Low to Moderate	0.0058 USD per Hour	0.0081 USD per Hour
	t2.micro	1	i386, x86_64	1	-	-	Low to Moderate	0.0116 USD per Hour	0.0162 USD per Hour
	t2.small	1	i386, x86_64	2	-	-	Low to Moderate	0.023 USD per Hour	0.032 USD per Hour
	t2.medium	2	i386, x86_64	4	-	-	Low to Moderate	0.0464 USD per Hour	0.0644 USD per Hour

To very large…and pricey!

	u-6tb1.56xlarge	224	x86_64	6144	-	-	100 Gigabit	46.40391 USD per Hour	56.70791 USD per Hour
	u-6tb1.112xlar…	448	x86_64	6144	-	-	100 Gigabit	54.6 USD per Hour	75.208 USD per Hour
	u-9tb1.112xlar…	448	x86_64	9216	-	-	100 Gigabit	81.9 USD per Hour	102.508 USD per Hour
	u-12tb1.112xl…	448	x86_64	12288	-	-	100 Gigabit	109.2 USD per Hour	129.808 USD per Hour

EC2 Instance Types

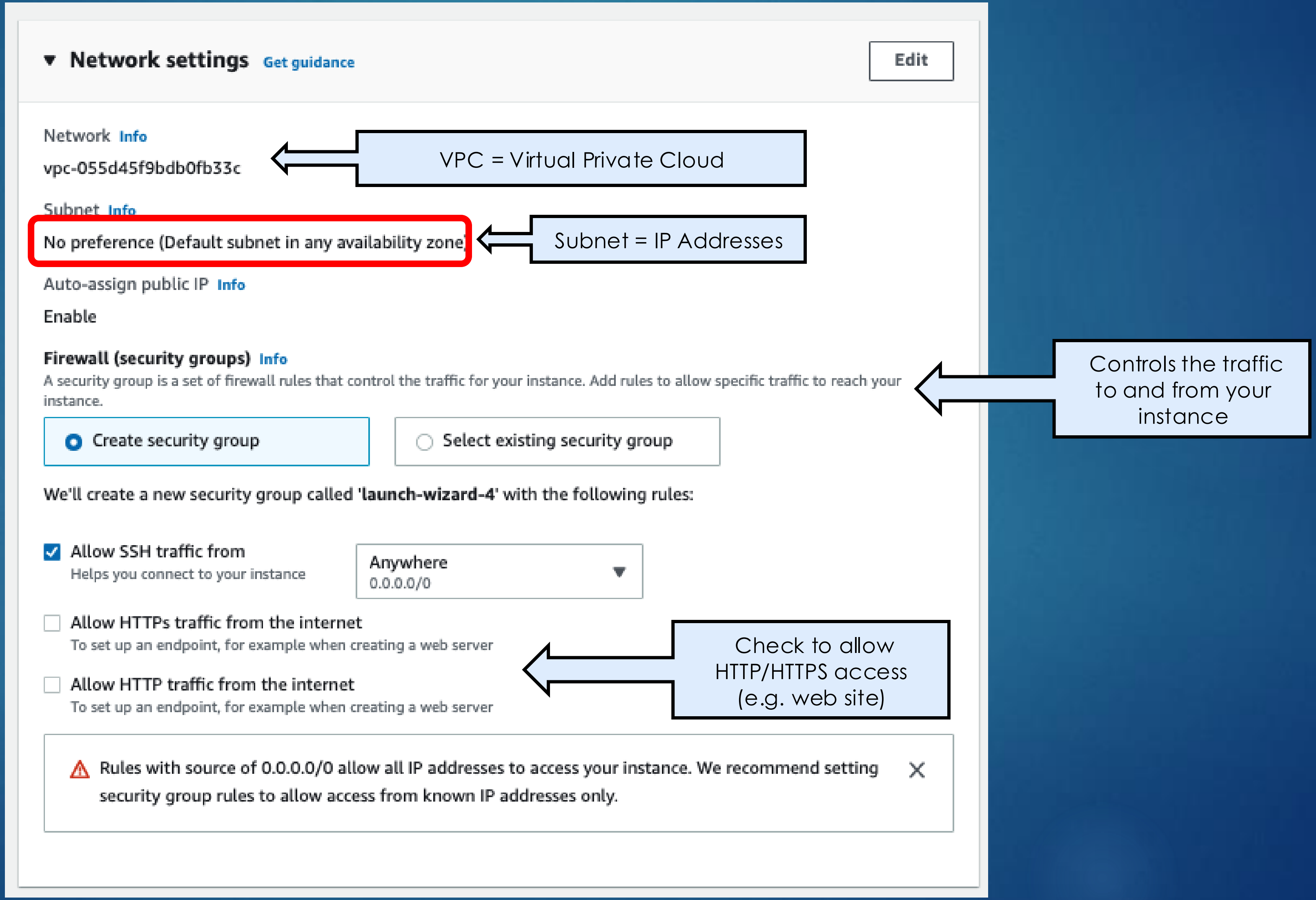
- Deciding which to use depends on what you need to do…

	Instance type	vCPUs	Architecture
○	t1.micro	1	i386, x86_64
○	t2.nano	1	i386, x86_64
●	t2.micro	1	i386, x86_64
○	t2.small	1	i386, x86_64
○	t2.medium	2	i386, x86_64

Family	Speciality	Use case
F1	Field Programmable Gate Array	Genomics research, financial analytics, real-time video processing, big data etc
I3	High Speed Storage	NoSQL DBs, Data Warehousing etc
G3	Graphics Intensive	Video Encoding/ 3D Application Streaming
H1	High Disk Throughput	MapReduce-based workloads, distributed file systems such as HDFS and MapR-FS
T2	Lowest Cost, General Purpose	Web Servers/Small DBs
D2	Dense Storage	Fileservers/Data Warehousing/Hadoop
R4	Memory Optimized	Memory Intensive Apps/DBs
M5	General Purpose	Application Servers
C5	Compute Optimized	CPU Intensive Apps/DBs
P3	Graphics/General Purpose GPU	Machine Learning, Bit Coin Mining etc
X1	Memory Optimized	SAP HANA/Apache Spark etc

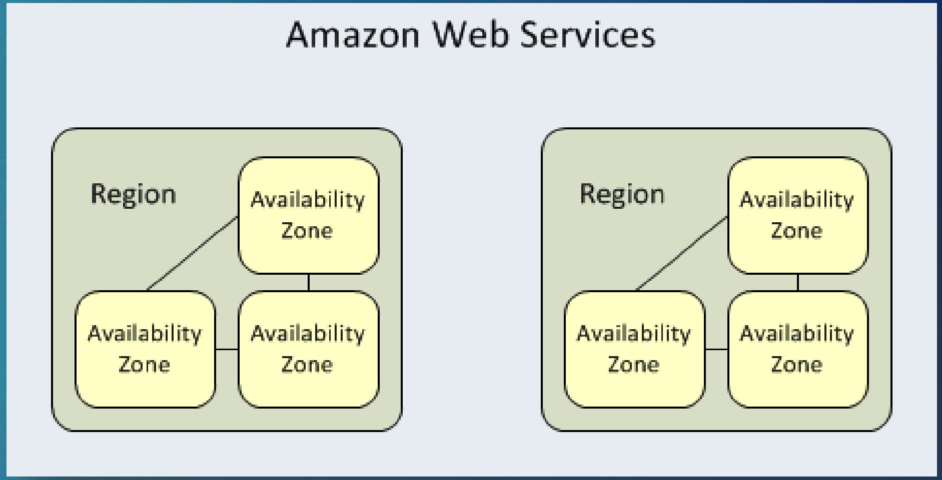
Source: https://medium.com/@priorww/acg-cliff-notes-ec2-101-a8dba0f71e0f

Network settings



Regions

- Each region represents a separate geographic area, where your EC2 instances are running
- It has multiple, isolated locations known as Availability Zones
- AWS resources (e.g. instances, key pairs, storage) are either:
  - Global
  - Tied to a Region
  - Tied to an Availability Zone

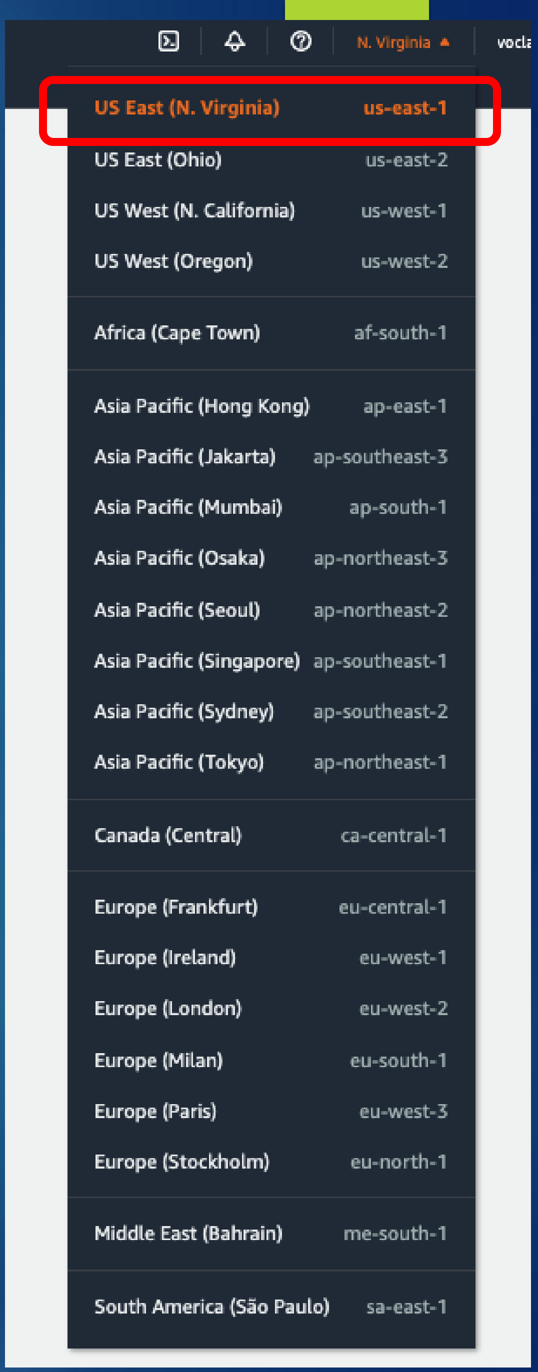


- Each Region is designed to be completely isolated from the other to provide fault tolerance and stability

Source: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Storage.html

Regions

- Why is this important?
  - Certain resources might not be available if created in different regions or availability zones
  - For example, your key pair is tied to the region it was created in
    - You cannot create an EC2 instance in another region with the same key pair unless you upload to that specific region
  - Another important factor is costs across regions
    - Any idea which is the most expensive?



Regions – US East vs South America

- Running the same EC2 instance in different regions can vary in costs

Region	Operating system
US East (N. Virginia)	Linux

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t4g.nano	$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	$0.0047	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.nano	$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t2.nano	$0.0058	1	0.5 GiB	EBS Only	Low
t4g.micro	$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	$0.0094	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.micro	$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit

Region	Operating system
South America (Sao Paulo)	Linux

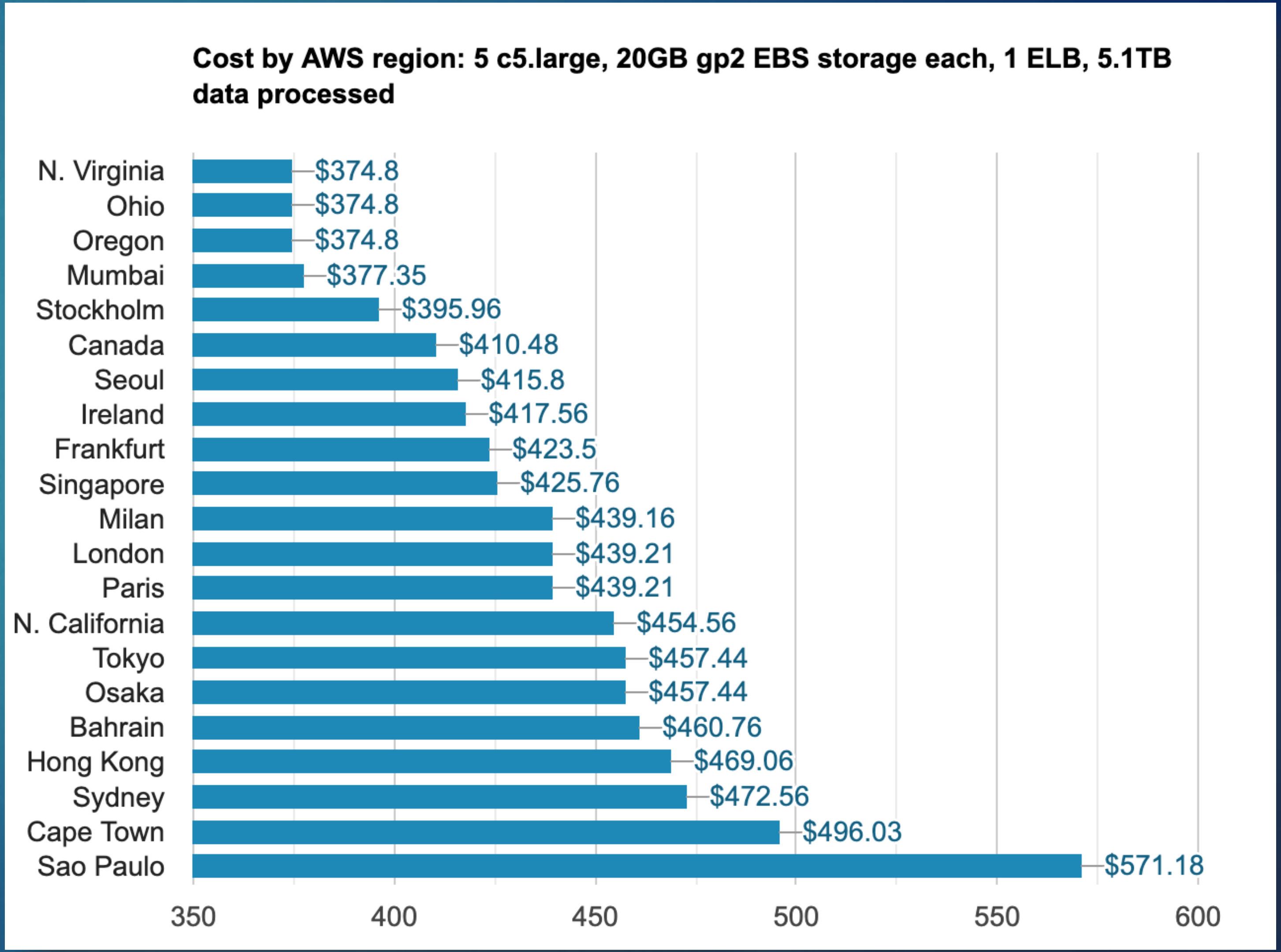
Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t4g.nano	$0.0067	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	$0.0076	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.nano	$0.0084	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t2.nano	$0.0093	1	0.5 GiB	EBS Only	Low
t4g.micro	$0.0134	2	1 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	$0.0151	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.micro	$0.0168	2	1 GiB	EBS Only	Up to 5 Gigabit

- The same rules for costs apply to storage and data transfer

Source: https://aws.amazon.com/ec2/pricing/on-demand/

Regions – Side by side comparison

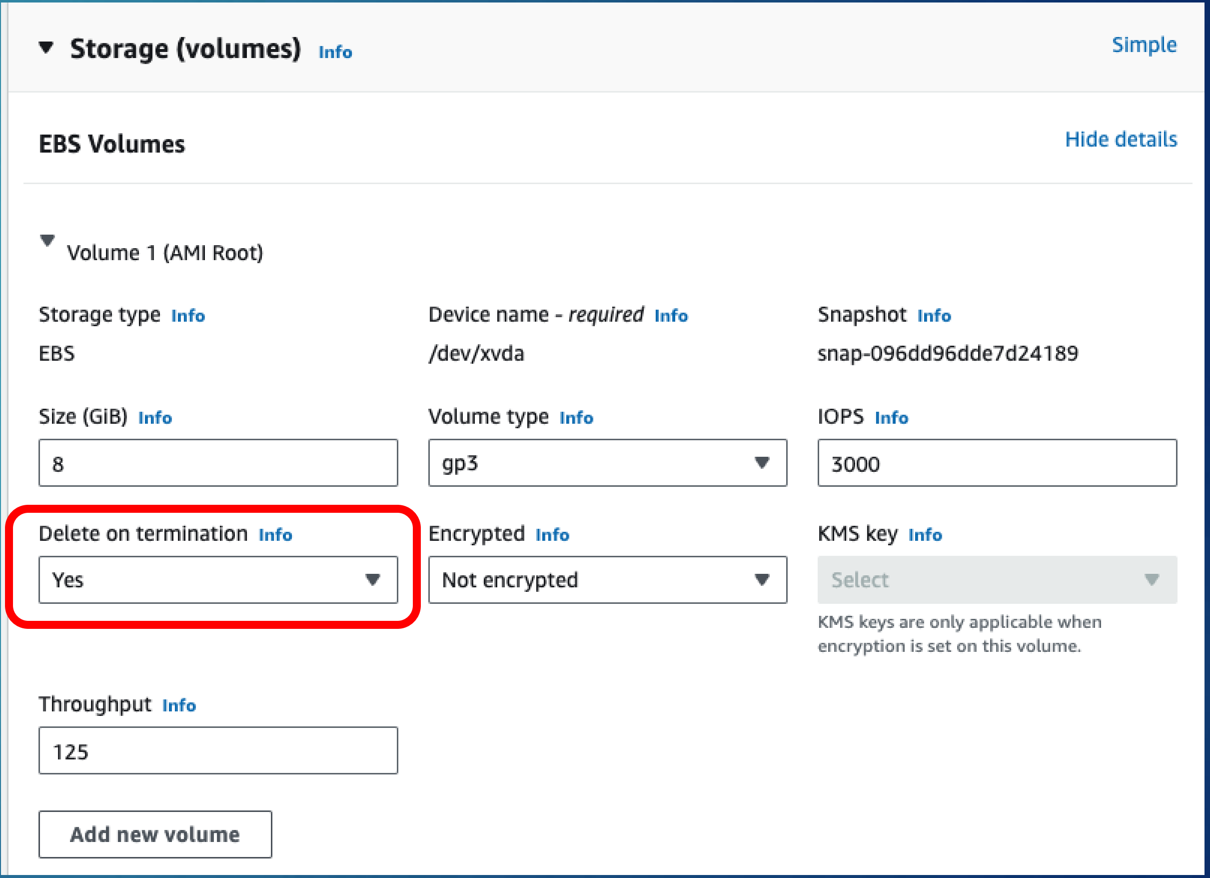
- Running the same hardware in Sao Paulo (Brazil) is **59% higher** than N. Virginia



Source: https://www.concurrencylabs.com/blog/choose-your-aws-region-wisely/

Storage

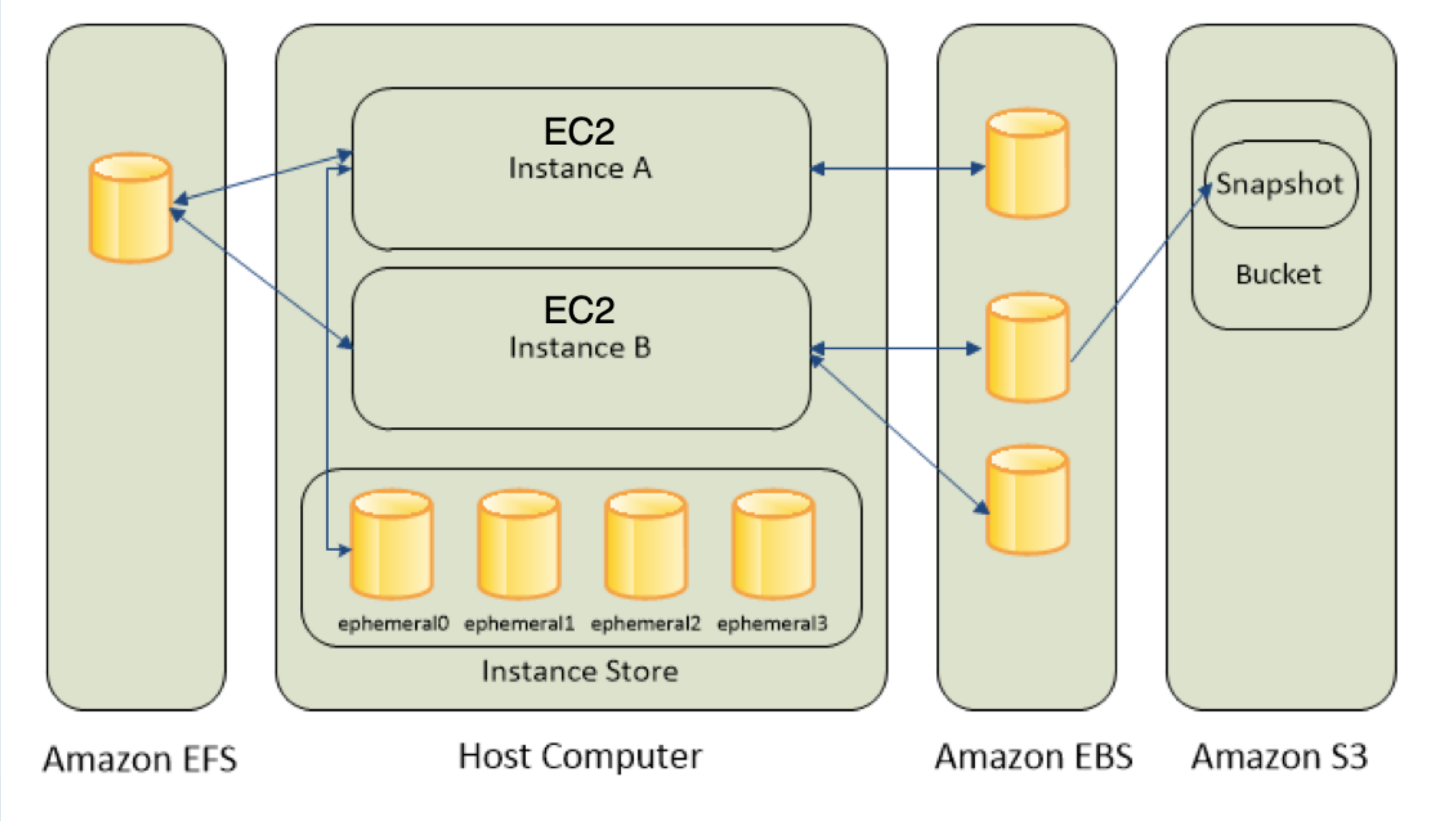
- Each instance has the option to add different types and sizes of storage
- EBS (Elastic Block Storage) volumes persist <u>independently</u> from the running life of EC2 instances
  - Why is this important?
- By default, EBS disk is deleted when the associated instance is terminated



Storage - Details

- Amazon EC2 offers 4 main storage options:
  1. Amazon EC2 Instance Store
  2. Amazon Simple Storage Service (S3)
  3. Amazon Elastic Block Store (EBS)
  4. Amazon Elastic File System (EFS)

Mounted to EC2 instance and exist independently



Source: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Storage.html

EC2 Instance Store

- Local to the instance only
- Provides very high random I/O performance, runs on SSD or HDD
- Non-persistent data store and data is lost under any of the following circumstances:
  - The underlying disk drive fails
  - The instance stops
  - The instance terminates
- Ideal for temporary storage of information that changes frequently, such as buffers, caches, scratch data, and other temporary content

Storage – Simple Storage Service (S3)

Amazon S3

- Provides simple object storage, useful for hosting simple websites with images and videos
- Every object stored in Amazon S3 is contained in a "bucket"
  - S3 buckets are created in a Region within a partition (group of regions)
    - There are 3: aws (Standard), aws-cn (China), and aws-us-gov (AWS GovCloud (US))
  - A bucket is similar to Internet domain names
  - AWS currently has three partitions: Objects stored in the buckets are retrieved using a HTTP URL address from anywhere
- Example: If an object with named "/photos/mygarden.jpg" is stored in the myawsbucket bucket, then it can be accessed by:

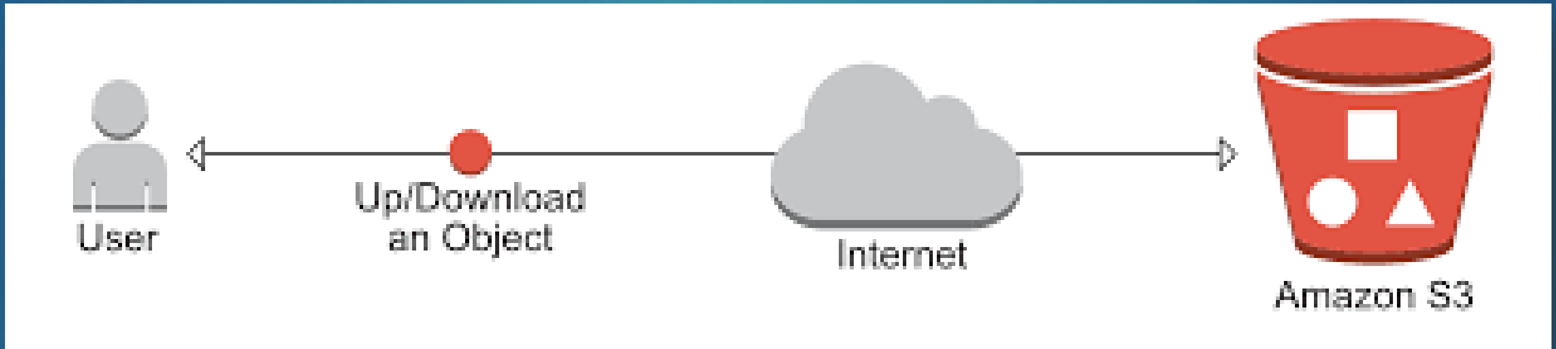
  https://myawsbucket.**s3.amazonaws.com**/photos/mygarden.jpg

  Bucket　　　　Object

- Question – What else is important about naming a bucket?

S3 – Try it for yourself – Part 1

- Do the exercise in Assignments > Activity – Store and Retrieve a file with S3
- Don’t forget to upload a screenshot when you are done



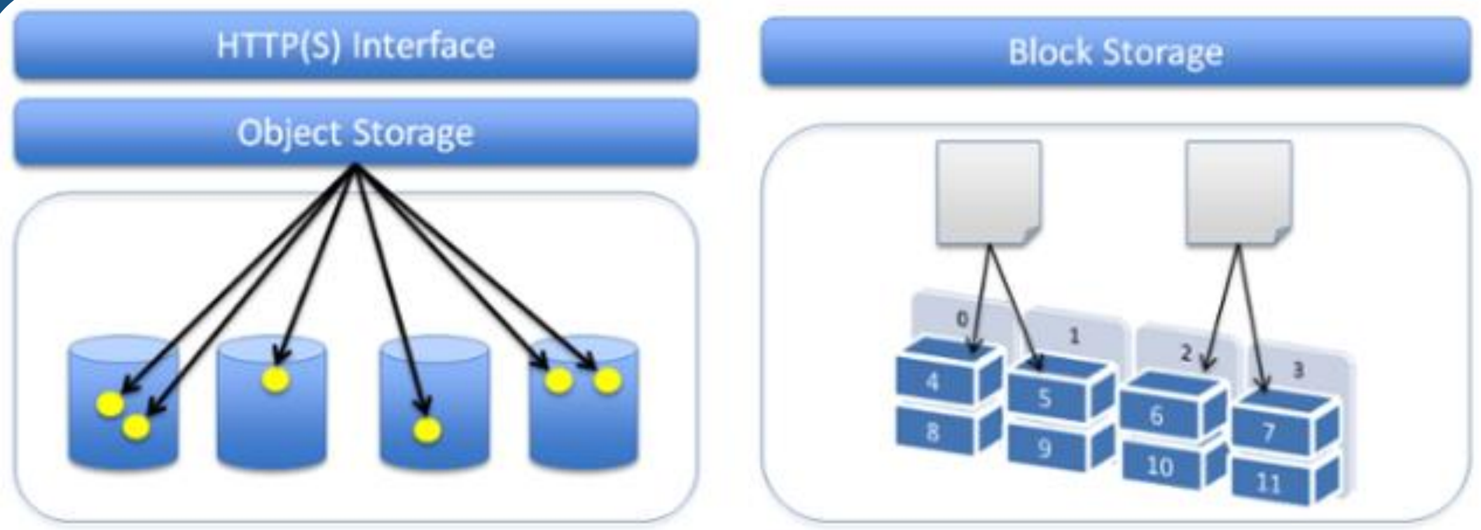
Storage – Elastic Block Store (EBS)

- Provides persistent block-level data storage
- A virtualized partition of a physical storage drive that's not directly connected to the EC2 instance it's associated with
- Block storage stores files in multiple volumes called "blocks", which act as separate hard drives
  - More flexible and offer higher performance than regular file storage
- Most EC2 instances are backed with EBS storage
- Use cases include business continuity, software testing, and database management

Storage – Elastic File System (EFS)

- Shared, elastic file storage system that grows and shrinks as you add and remove files
- It offers a traditional file storage paradigm, with data organized into directories and subdirectories
- EFS is useful for SaaS applications and content management systems
- You can mount EFS onto several EC2 instances at the same time

Storage – Costs/Comparisons



S3

- Store virtually unlimited files.
- Maintain file revisions.
- HTTP(S) based interface.
- Files are distributed in different physical nodes.

EBS, EFS

- File is split and stored in fixed sized blocks.
- Capacity can be increased by adding more nodes.
- Suitable for applications which require high IOPS, database, transactional data.

Source: https://dzone.com/articles/confused-by-aws-storage-options-s3-ebs-amp-efs-explained

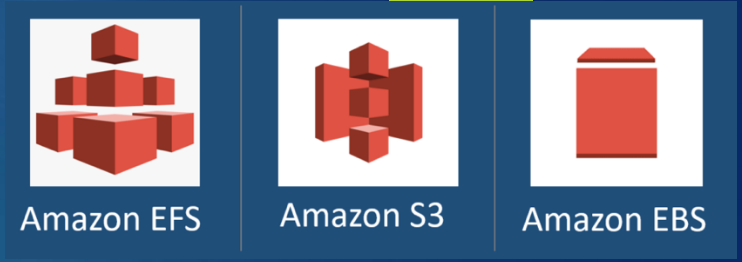
EBS vs. EFS vs. S3

- Your free tier account comes with:
  - EBS = 30GB / 12 months free
  - EFS = 5GB / 12 months free
  - S3 = 5GB / 12 months free

Feature	Amazon EBS	Amazon EFS	Amazon S3
Type of Storage	Block Storage	File Storage	Object Storage
Use Cases	EC2 instance storage, databases	Shared file storage, content serving	Data lakes, backup, archival, websites
Performance	Volume types with varying IOPS	Scales automatically	Highly scalable, different storage classes
Durability & Redundancy	High within a single AZ	High across multiple AZs	High across multiple data centers
Access	Attached to a single EC2 instance	Shared across multiple EC2 instances	Accessible via HTTP/HTTPS, versatile access patterns
Scaling	Vertical scaling	Horizontal scaling	Highly scalable, automatic scaling based on demand
Latency	Low	Low	Variable, designed for web-scale traffic
Cost Structure	Pay for provisioned capacity	Pay for storage used	Pay for storage, requests, and data transfer
Snapshot Capability	Yes	No	Versioning and object lifecycle management
Integration with AWS Services	Integrated with EC2	Integrates with EC2 and on-premises servers	Integrates with various AWS services, widely used in cloud architectures

Source: https://medium.com/@mohitgadhavi1/aws-storage-choosing-between-ebs-efs-and-s3-for-optimal-solutions-8e213a900e56

Storage – So what do you use?



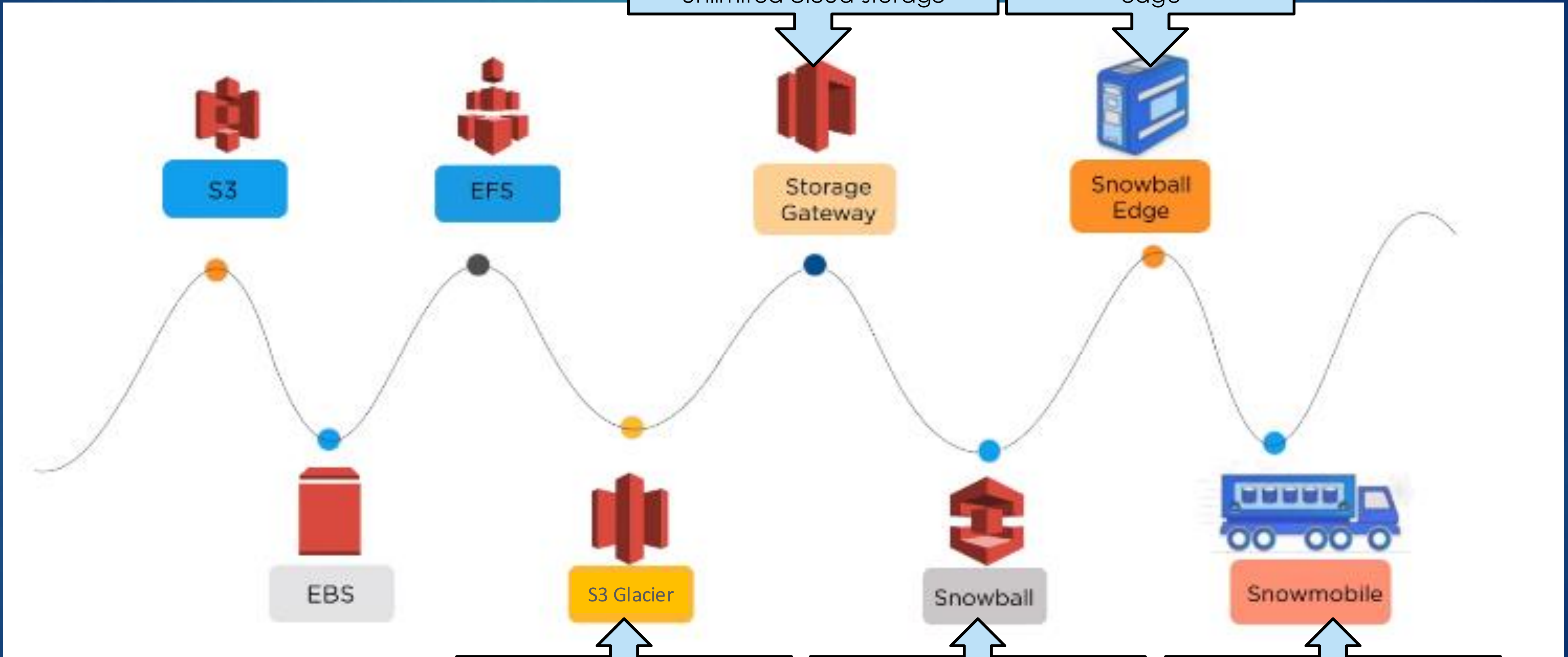
- Simple Storage Service (S3)
  - Cheapest for data storage alone and can be accessed from anywhere via a URL
  - Highly Available and redundant. 99.999999999% durability
- Elastic Block Store (EBS)
  - Faster than S3 and EFS, more expensive than S3 but less costly than EFS
  - Only available in a <u>same region</u> as the EC2 instance
- Elastic File System (EFS)
  - Faster than S3, most expensive compared to S3 and EBS
  - Available on <u>multiple regions</u> and can be accessed by *thousands* on EC2 instances at once
  - Best for large very large quantities of data, such as large analytic workloads, which may exceed the capabilities of EBS

Conclusion: A Balance of <u>Cost</u> and <u>Necessity</u>

Storage – Alternative options

Hybrid model provides on-premise access to virtually unlimited cloud storage

Petabyte compute and storage on the edge



Very low cost storage (1$ per TB/month)

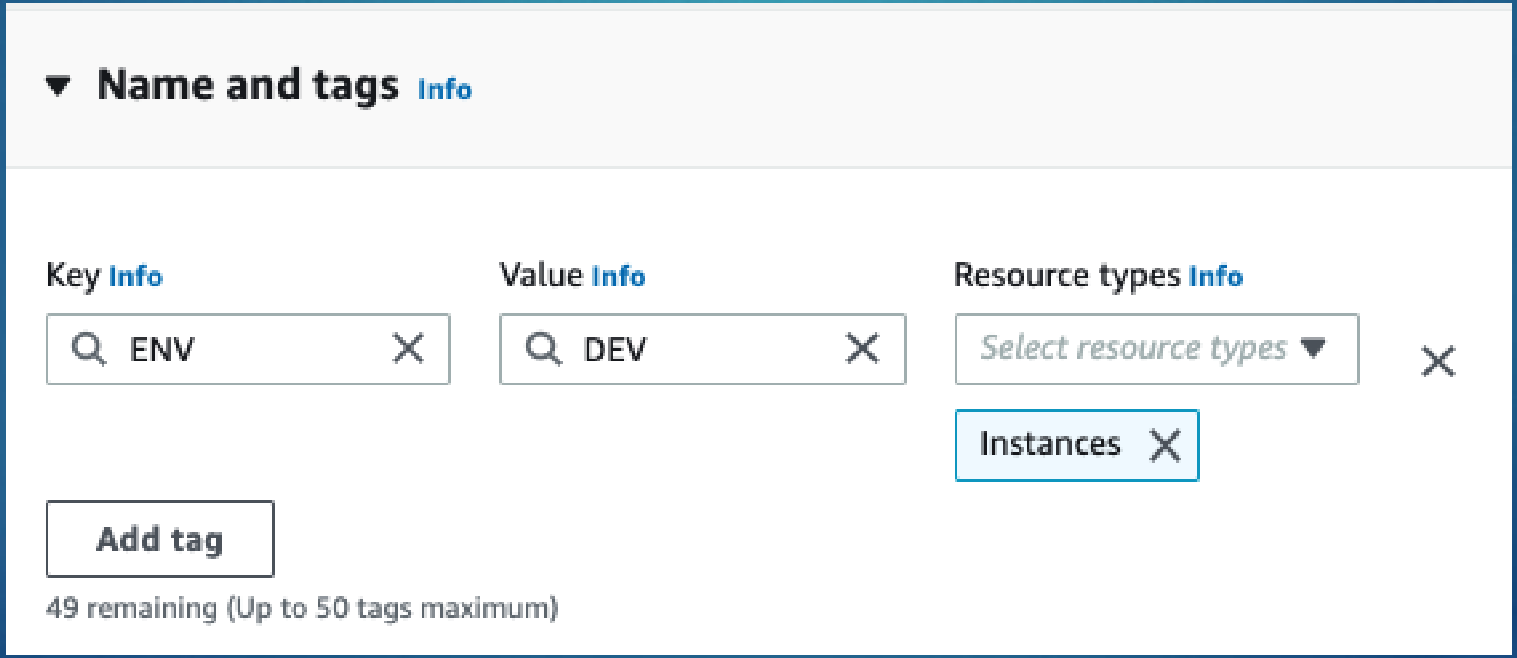
Petabyte transfer by secure appliance

Exabyte-scale (Up to 100PB) transfer by truck

Source: https://www.simplilearn.com/tutorials/aws-tutorial/aws-s3

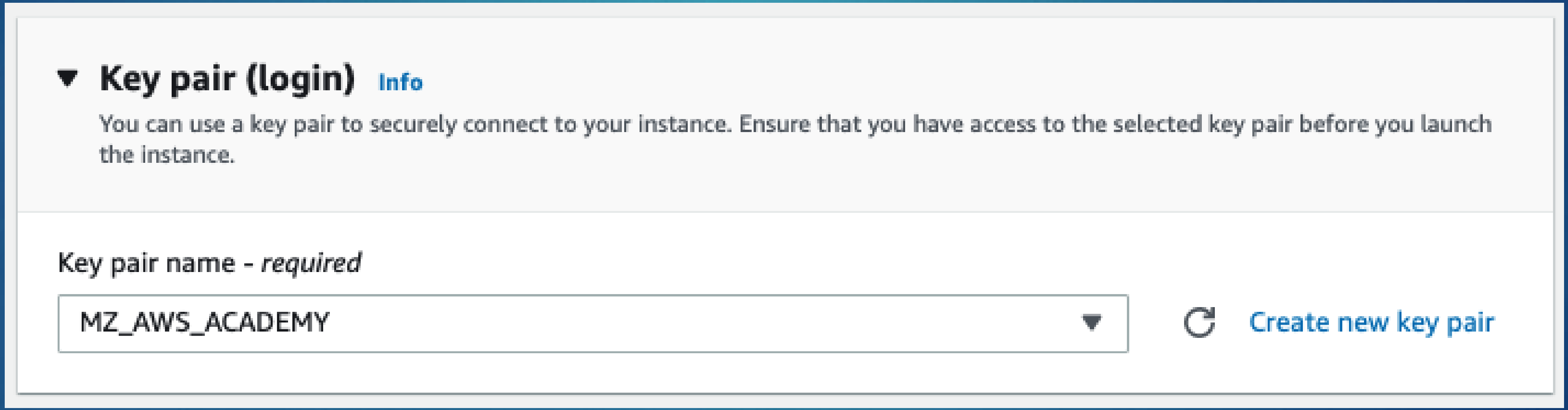
Tagging

- Each tag is a simple label consisting of a user-defined key and an optional value
- Makes it easier to manage, search for, and filter resources for billing, environments (dev/test/prod), etc.



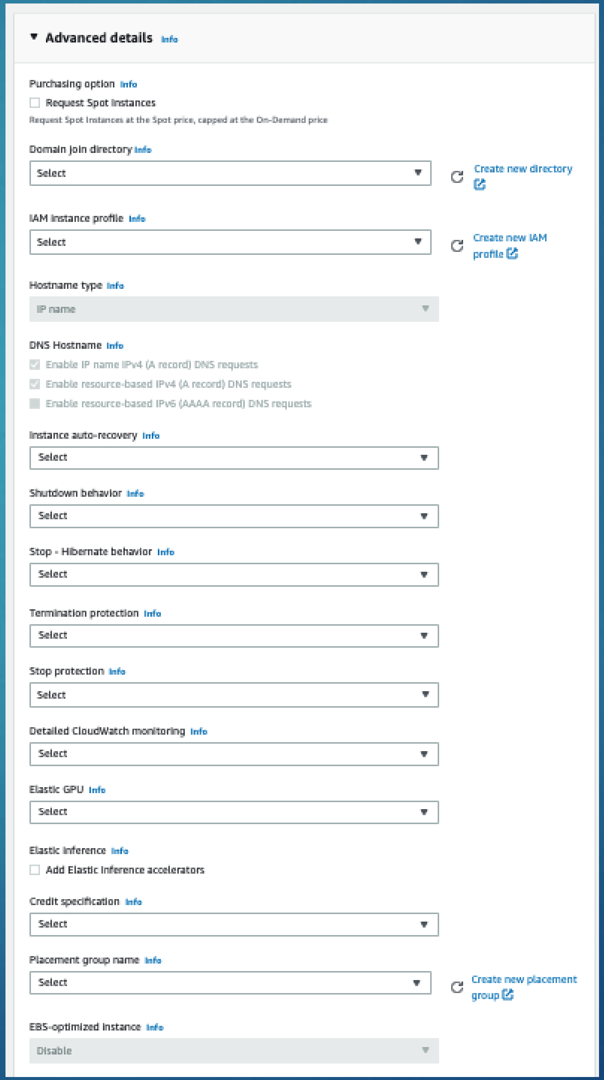
Specify Key Pair

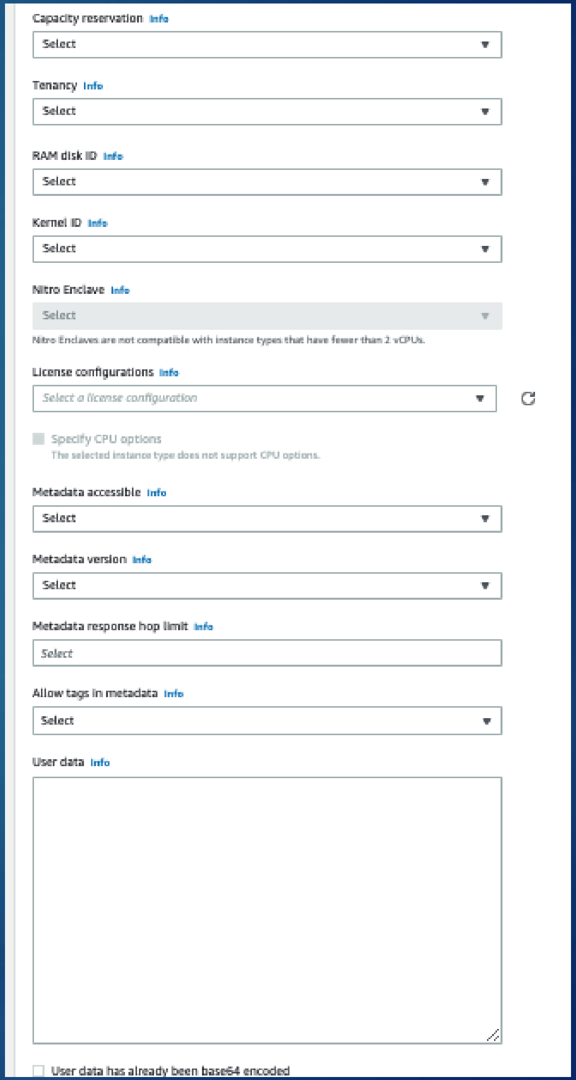
- Required for accessing your EC2 instance and other AWS resources
- NOTE: If you lose your key pair or run into another other issues, you can easily create a new pair under EC2 > Network & Security > Key Pairs



Advanced Options

- There are many more options under “Advanced details” to further configure your EC2 instances
- We will revisit this later

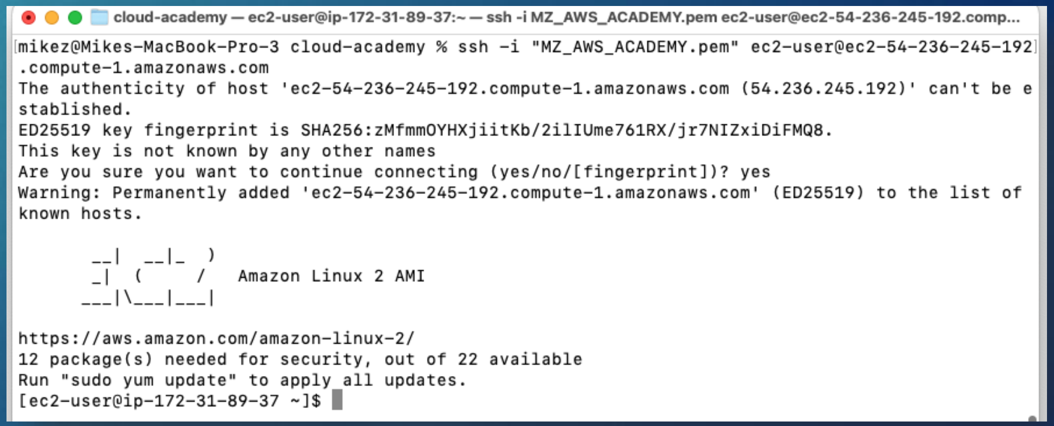




Accessing EC2: CloudShell vs. your PC

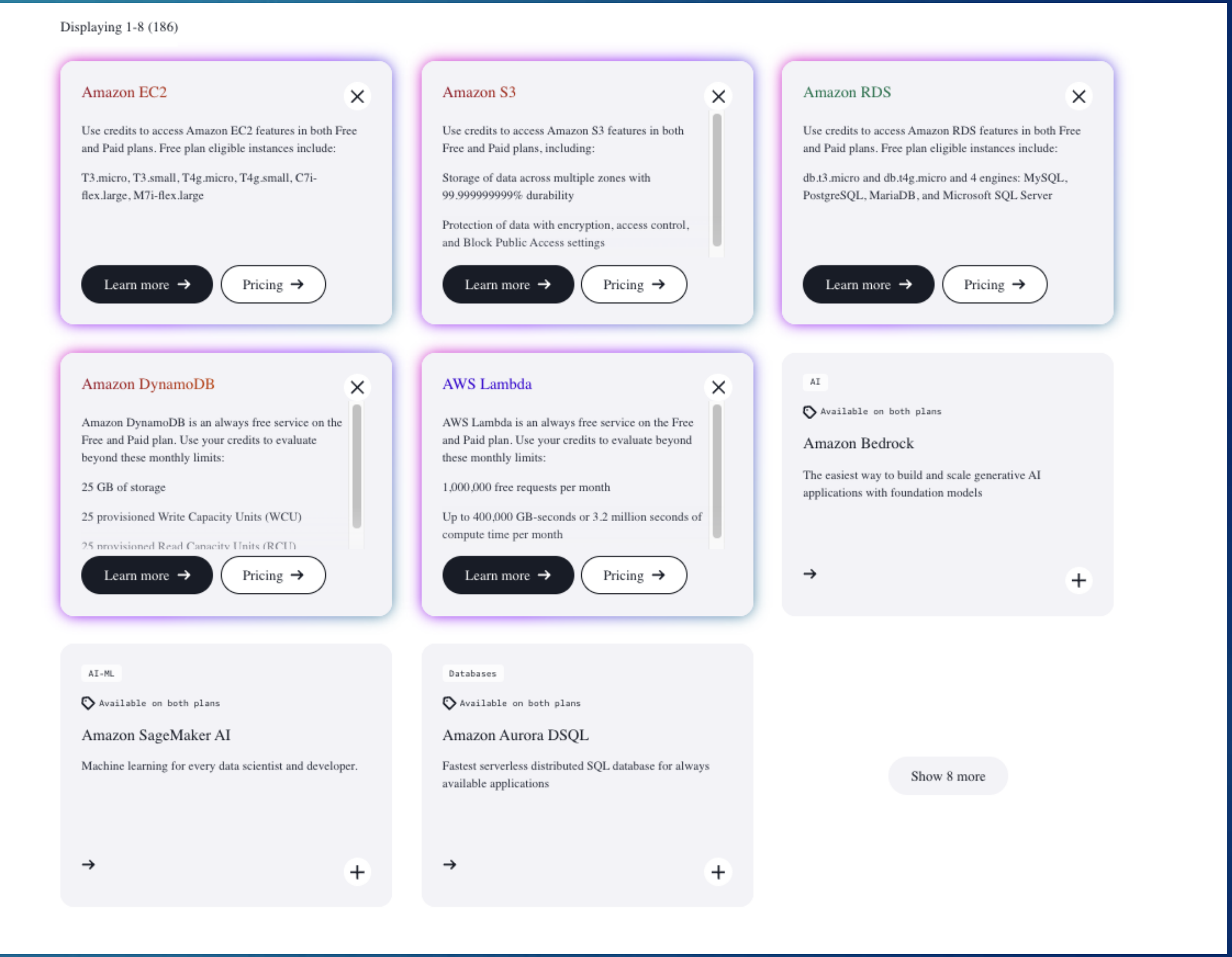
- CloudShell runs from your browser, and you can do use this for most activities and term project. It comes with 1GB per region. Disk is persisted, installed software is not.

- If you prefer using your PC, there are a lot of options (Git bash, PuTTY, powershell, etc.)
  - Some easier than others to set up
- You will need to install **AWS CLI** (Command-line Interface) to access AWS
- AWS CLI is pre-installed on CloudShell



AWS Accounts

- If you are using a new AWS service, make sure you check whether how it's covered under your account (see right)
- There will be some activities we will do this semester that will require a slight charge



Where we are now…

- You are now able to create an EC2 instance and your own S3 bucket
- Let's try something more interesting using your S3 bucket
- You next assignment is to create a public static web site with your S3 bucket to make it accessible to anyone
- Go to myCourses Assignments > Activity – Create a Static Web Site with S3

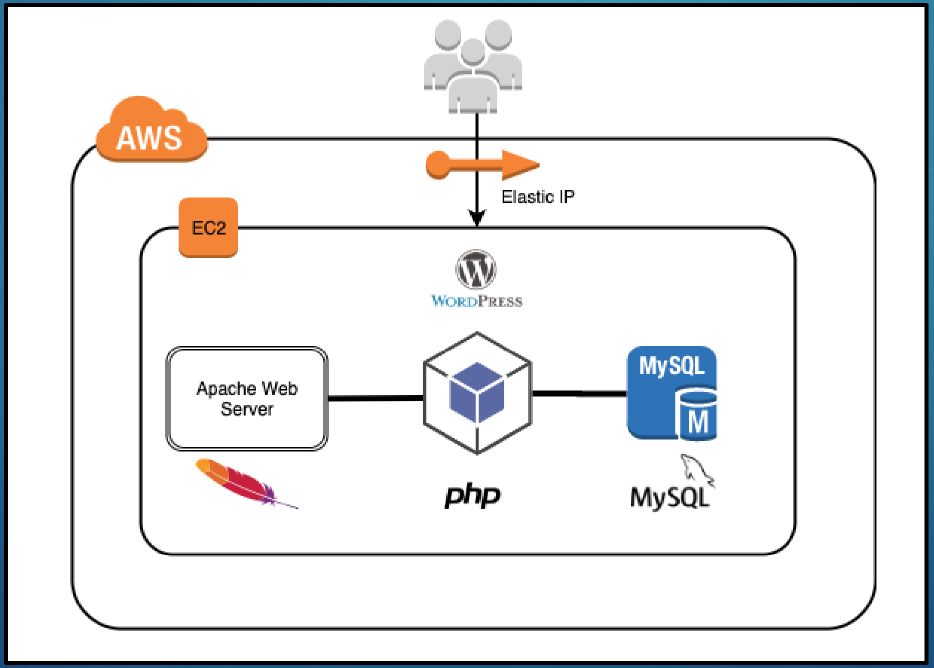
Term Project Teams

- There will be 1 private channel per team on Slack that I am asking teams use for their communication
  - Look for invites very shortly
- This will help me, and the CAs monitor team progress and quickly answer any questions you may have for us

- Homework – AWS Tutorials (3 points each)
- Each team member picks and completes two AWS tutorials
  - Purpose of this is to get your team more familiar with AWS services for the term project
  - Team members can do a tutorial on the same AWS technology, but these must be different tutorials (make sure you coordinate with your team!)
- Details and guidelines can be found in myCourses under Assignments > Homework - AWS Tutorials
  - All submissions should be posted to Discussions not Assignments
  - These will only be visible to your team, but eventually to the entire class

Due next class:
Assignments > Activity - Install WordPress on EC2

- **WordPress** is a content management system (CMS) that allows you to create and manage websites
- It's the world's most popular website builder and powers over 43% of all websites



- Make sure you allocate enough time for this activity (20-30 minutes)
- Read the directions carefully!
- Do not do it right before next class